

NBT stain

Detect superoxide. Form insoluble dark-blue formazan precipitate after reduced by $O_2^{\bullet-}$.

[Invitrogen \(N6495 \)](#), [Santa Cruz \(sc-296003A \)](#)

Nitro blue Tetrazolium Chloride

1 mg / mL **NBT powder**
10 mM **sodium phosphate buffer (pH 7.8)**
10 mM **NaN₃ (sodium azide)**

NBT working solution

Dissolve the NBT powder in [sodium phosphate buffer](#). The working solution must be **protected from light**.

Chemical	M.W. (g / mol)	Concentration	Addition (mg)	Location
NaH ₂ PO ₄	119.98	1.342 mM	161.0	III
Na ₂ HPO ₄	141.96	8.657 mM	1228.0	III
NaN ₃	65.01	10 mM	650.1	IV
NBT powder	817.65	1 mg / mL	1000.0	VII
ddH ₂ O			1 L	

- The working solution should be better filtered before use. The quality among batches is not consistent. Some batches can be fully dissolved in the buffer, resulting clear yellowish working solution (the staining result is dark blue). However, some batches will form reddish brown precipitates (the staining result will be reddish-darkbrown or greyish-darkblue, and tend to be over-stained).

Procedure

1. Directly immersed the root into the NBT solution (pre-warmed to 30°C), avoid the roots from touching the tube wall.
 2. Vacuum 20 minutes under ~ 650 mmHg (the lid should be opened).
 3. (optional) Close the lid and incubate under 30°C for 10 minutes ~ 2 hours, depend on the root condition.
- Generally, 10 mins is appropriate for young root (less than one week old), and the extremely old root (more than two weeks old) may require 2 hours.

Done !!!
